

Total No. of Questions : 8]

SEAT No. :

PD4074

[Total No. of Pages : 3

[6402]-34

S.E. (Electronics and Computer Engineering)
SYSTEM PROGRAMMING AND OPERATING SYSTEMS
(2019 Pattern) (Semester - IV) (204207)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What is an operating system? What are the goals or objectives of operating system? [6]

b) Consider the following processes arrival time and burst time are as shown. Calculate average waiting time and average turnaround time using FCFS scheduling algorithm. [6]

Process	Arrival time	Burst Time
P1	2	6
P2	1	8
P3	0	3
P4	4	4

c) Draw and explain architecture of Batch Operating Systems. Write the advantages and disadvantages of it. [8]

OR

Q2) a) Explain Round Robin(RR) Scheduling, State advantages and disadvantages. [6]

b) In the following example, there are four processes named as P1, P2, P3, and P4. Their arrival time as 0 for all and burst time are given in the table below. Calculate average TAT and WT for SJF (Non-preemptive) scheduling Algorithm. [6]

Process	Burst Time
P1	21
P2	3
P3	6
P4	2

c) Draw and explain Monolithic Kernel operating system. Differentiate between Microkernel and Monolithic Kernel. [8]

P.T.O.

- Q3)** a) Explain the terms Shared Memory and Message passing for IPC. Give the difference between shared memory and message passing. [8]
b) What is Process synchronization? Explain Critical Section in detail. Explain various solutions the Critical Section problem in brief. [8]

OR

- Q4)** a) What is dead lock in operating system? Explain the conditions under which deadlock occur? Differentiate between Starvation and Deadlock. [8]
b) What is semaphore? Explain different types of Semaphore. State the advantages and disadvantages of Semaphore. [9]

- Q5)** a) Explain contiguous memory allocation and non-contiguous memory allocation. Write the difference between them. [8]
b) Explain different types of memory fragmentations? Differentiate them on following points. [8]
i) Definition
ii) Occurrence
iii) Solution

OR

- Q6)** a) Explain concept of paging with suitable diagram. Give advantages & disadvantages of the paging mechanism. [8]
b) Consider a reference string: 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. The number of frames is 3. Find out the number of page faults respective to:
i) FIFO Page Replacement Algorithm
ii) LRU Page Replacement Algorithm

- Q7)** a) Consider the following disk request sequence for a disk with 100 tracks 43, 52, 24, 65, 70, 48, 16, 61. Head pointer is starting at 20 and moving in left direction. Find the number of head movements (total seek length) in cylinders using FCFS scheduling. [6]
b) Differentiate between Programmed I/O Transfer and DMA I/O Transfer. [6]
c) Explain the following file organization techniques [6]
i) Sequential file organization
ii) Random or direct file organization

OR

- Q8)** a) What is I/O buffering? Explain circular buffering and state its advantages and disadvantages. [6]
- b) Explain the following File allocation methods. [6]
- i) Contiguous File allocation
 - ii) Linked File Allocation.
- c) Explain disk scheduling concept. Why is disk scheduling needed? Explain any one disk scheduling algorithms in detail. [6]

